

Part II (35 marks)

C. Answer all these questions.

16. Draw a typical pressure distribution in the contact surface of a rolling process as a function of contact angle. [2]

17. What are the basic differences between upset forging and swaging? [2]

18. What are the typical measures to avoid hot cracking? [2]

19. Determine the solidification time for iron casting of the following cases when molten iron is poured into sand mould at an initial temperature of 25°C . [3+3+3]

(a) a slab-shaped casting of 8 cm thick

(b) a sphere of 8 cm diameter

(c) an infinitely-long cylinder of 8 cm diameter

Use the following data for iron: freezing temperature 1540°C , latent heat 272 kJ/kg , density 7850 kg/m^3 , and for sand: specific heat 1.17 kJ/kg-K , thermal conductivity 0.9 W/m-K and density 1600 kg/m^3 .

20. A strip of lead with initial dimensions $24 \text{ mm} \times 24 \text{ mm} \times 150 \text{ mm}$ is forged between flat dies to a final size of $6 \text{ mm} \times 96 \text{ mm} \times 150 \text{ mm}$. If the coefficient of friction between the job and the dies is 0.25, determine the maximum forging force. The average yield stress of lead in tension is 7 N/mm^2 .

Note: Derive the mathematical expression at first and then solve the problem. Assume von Mises criterion for yielding. [9]

21. (a) Derive the mathematical expression of total spring back in terms of bending moment, geometric parameters and materials properties for a sheet-metal bending process. Use neat sketches for the derivation.

(b) What is the required punch angle to produce a bend with a given included angle? [5+1]

22. Derive the mathematical expression of the mould filling time for a bottom gating system in terms of geometric parameters of the system. Use a neat sketch for the derivation. [5]